

Problem 1

Let I be an ideal in the ring R , and J be an ideal in the ring S . Consider $I \times J$.

Let $(i,j), (a,b) \in I \times J$. Consider:

$$(i,j) + (a,b) = (i+a, j+b)$$

Since I and J are ideals this implies that $i+a \in I$ and $j+b \in J$. Thus, $(i+a, j+b) \in I \times J$

Let $(r,s) \in R \times S$ and $(i,j) \in I \times J$. Consider:

$$(r,s)(i,j) = (ri, sj)$$

Since I and J are ideals this implies that $ri \in I$ and $sj \in J$. Thus, $(ri, sj) \in I \times J$

Thus $I \times J$ is an ideal.

Let K be an ideal in $R \times S$. Consider:

$$I = \{r \in R \mid (r,0) \in K\}$$

$$J = \{s \in S \mid (0,s) \in K\}$$

Easy to show that these are ideals. Let $a, b \in I$ then $(a,0), (b,0) \in K \implies (a+b,0) \in K$ thus $a+b \in I$

Let $r \in R, i \in I$ then $(i,0) \in K$ thus $(ri,0) \in K$ thus $ri \in I$ same thing works for J .

Let $(i,j) \in I \times J$ thus $(i,0), (0,j) \in K$ thus $(i,0) + (0,j) = (i,j) \in K$.

Let $(r,s) \in K$. Multiply by $(1,0), (0,1)$ thus we get that $(r,0) \in K$, and $(0,s) \in K$. Thus, $r \in I$ and $s \in J$ thus $(r,s) \in I \times J$. Thus, $K = I \times J$ so yes every ideal is in that form.

Problem 2

Assume I and J are coprime. Thus, $I + J = (1)$. Thus, there is $i \in I$ and $j \in J$ s.t. $i + j = 1$.

Let $a \in IJ$ by 2(a) we know that $IJ \subset I \cap J$ thus $a \in I \cap J$

Let $a \in I \cap J$. Thus, $ai + aj = a$. Thus, a can be written as a finite sum of product of elements in I, J thus $a \in IJ$. Thus, $IJ = I \cap J$.

In $\mathbb{Z}$ this is equivalent to saying if a, b are coprime then the least common multiple is equal to ab

Problem 3

Assume I, J are ideals in R .

- Equality of $r(I \cap J)$ and $r(IJ)$:

Let $x \in r(I \cap J)$. Thus, $x^n \in I \cap J$ for some $n \in \mathbb{N}$. Since $x^n \in I \cap J$, we know that $x^n \cdot x^n = x^{2n} \in IJ$, as this is a finite sum of products of elements from I and J .

Conversely, let $x \in r(IJ)$. Thus, $x^n \in IJ$ for some $n \in \mathbb{N}$. By part 2(a), we know $x^n \in I \cap J$, which implies $x \in r(I \cap J)$.

Hence, $r(I \cap J) = r(IJ)$.

- Equality of $r(I \cap J)$ and $r(I) \cap r(J)$:

Let $x \in r(I) \cap r(J)$. Then there exist $k, n \in \mathbb{N}$ such that $x^k \in I$ and $x^n \in J$. This implies $x^{k+n} \in I \cap J$, so $x \in r(I \cap J)$.

Conversely, let $x \in r(I \cap J)$. Then $x^n \in I \cap J$ for some $n \in \mathbb{N}$. Thus, $x^n \in I$ and $x^n \in J$, which implies $x \in r(I)$ and $x \in r(J)$. Hence, $x \in r(I) \cap r(J)$.

Therefore, $r(I \cap J) = r(I) \cap r(J)$.

- Equality of $r(I + J)$ and $r(r(I) + r(J))$:

Let $x \in r(r(I) + r(J))$. Then $x^n \in r(I) + r(J)$ for some $n \in \mathbb{N}$. Thus, $x^n = a + b$ for some $a \in r(I)$ and $b \in r(J)$. Since $a \in r(I)$, there exists $j \in \mathbb{N}$ such that $a^j \in I$, and since $b \in r(J)$, there exists $k \in \mathbb{N}$ such that $b^k \in J$. Consider

$$x^{n^{j+k}} = (a + b)^{j+k}.$$

By the binomial theorem, every term in this expansion will involve either a^d or b^c with $d \geq j$ and $c \geq k$. Since $a^d \in I$ and $b^c \in J$, every term is in $I + J$. Therefore, $x^{n^{j+k}} \in I + J$, so $x \in r(I + J)$.

Conversely, let $x \in r(I + J)$. Then $x^n \in I + J$ for some $n \in \mathbb{N}$. Thus, $x^n = a + b$ for some $a \in I$ and $b \in J$. Since $a \in I$, $a \in r(I)$, and since $b \in J$, $b \in r(J)$. Thus, $x^n \in r(I) + r(J)$, so $x \in r(r(I) + r(J))$.

Therefore, $r(I + J) = r(r(I) + r(J))$.

- Combining the results above, we have:

$$r(I \cap J) = r(IJ) = r(I) \cap r(J),$$

and

$$r(I + J) = r(r(I) + r(J)).$$

Problem 4

(a) Let a be nilpotent with $a^n = 0$ for $n > 0$. Consider $1 + a$. Thus:

$$(a + 1)(1 - a + a^2 + \cdots + (-1)^{n-1}a^{n-1}) = 1$$

Thus $1 + a$ is a unit. Thus if u is a unit then $u^{-1}a$ is nilpotent. Thus $1 + u^{-1}a$ is a unit, thus $u(1 + u^{-1}a) = u + 1$ is a unit as well.

Assume a_0 is a unit and the rest of the terms are nilpotent then it is easy to see that $a_1x, a_2x^2, \dots, a_nx^n$ are also nilpotent. Thus since things that are nilpotent are closed under addition we get that $a_1x + \cdots + a_nx^n = A$ is nilpotent. Thus by the argument above $a_0 + A$ is a unit. Thus f is a unit.

Assume f is a unit in $R[X]$. Thus, there is a function g s.t. $fg = 1$.

Base Case:

Let $f = a_0$ well if f is a unit then since it is degree zero a_0 must also be a unit.

Inductive Step:

Assume for any polynomial of degree $n - 1$ that is a unit has a unit constant term and the rest of the terms of nilpotent.

Consider:

$$fg = (a_0 + a_1x + \cdots + x^n)(b_0 + b_1x + \cdots + b_mx^m) = 1$$

It is easy to see that $a_0b_0 = 1$ thus a_0 is a unit.

It is also easy to see that $a_nb_m = 0$. Then consider the x^{n+m-1} term we can see that $a_{n-1}b_m + b_{m-1}a_n = 0$. Multiplying by a_n gives $b_{m-1}a_n^2 = 0$. Then for the x^{n+m-2} term we get $a_{n-2}b_m + a_{n-1}b_{m-1} + a_nb_{m-2} = 0$. Multiplying by a_n^2 gives $b_{m-2}a_n^3 = 0$. Continue this patten till you get $b_0a_n^{m+1} = 0$. Then since b_0 is a unit we can cancel it and get $a_n^{m+1} = 0$. Thus, it is nilpotent.

Now consider $f(x) - a_nx^n$. From the proof above since a_nx^n is nilpotent and $f(x)$ is a unit $f(x) - a_nx^n$ is a unit of degree less than n . Thus by the inductive hypothesis we get that all the terms in $f(x) - a_nx^n$ are nilpotent.

(b) Assume $a_0, a_1, \dots, a_n$ is nilpotent. Thus it is easy to see that $a_0, a_1x, \dots, a_nx^n$ are also nilpotent. Thus since things that are nilpotent are closed under addition we know that $f(x) = a_0 + \cdots + a_nx^n$ is nilpotent.

Assume f is nilpotent. Thus, $f^k = 0$ for some $k > 0$ Thus, we know that xf is also nilpotent. Then $1 + xf$ is a unit by the proof above. Thus, by (a) we know all the terms except the constant one are nilpotent thus we are done.

Problem 5

(a) **(Forward direction):** Assume f is a unit in $R[[x]]$. Thus, there exists a power series g such that $fg = 1$. Since the constant term of the product fg is $a_0b_0 = 1$, it follows that a_0 is a unit in R .

(Converse direction): Assume a_0 is a unit in R . Let $f = a_0 + a_1x + a_2x^2 + \dots$ and define $g = b_0 + b_1x + b_2x^2 + \dots$. We aim to construct g such that $fg = 1$.

Expanding the product fg , we have:

$$fg = a_0b_0 + \sum_{n=1}^{\infty} \left(\sum_{i=0}^n a_i b_{n-i} \right) x^n.$$

We need $fg = 1$, which means the constant term satisfies $a_0b_0 = 1$, and all higher-order terms must vanish, i.e.,

$$\sum_{i=0}^n a_i b_{n-i} = 0 \quad \text{for } n \geq 1.$$

Let $b_0 = a_0^{-1}$. For $n \geq 1$, recursively define:

$$b_n = -a_0^{-1} \sum_{i=1}^n a_i b_{n-i}.$$

Base Case: The constant term satisfies $a_0b_0 = 1$, which holds by definition of $b_0 = a_0^{-1}$.

Inductive Step: Assume the formula for b_k holds and makes the x^k term zero for all $k < n$. Consider the x^n term in fg :

$$\sum_{i=0}^n a_i b_{n-i} = a_0 b_n + \sum_{i=1}^n a_i b_{n-i}.$$

Substituting $b_n = -a_0^{-1} \sum_{i=1}^n a_i b_{n-i}$ into the equation:

$$a_0 b_n + \sum_{i=1}^n a_i b_{n-i} = a_0 \left(-a_0^{-1} \sum_{i=1}^n a_i b_{n-i} \right) + \sum_{i=1}^n a_i b_{n-i}.$$

Simplifying:

$$-a_0 \cdot a_0^{-1} \sum_{i=1}^n a_i b_{n-i} + \sum_{i=1}^n a_i b_{n-i} = 0.$$

Thus, the inductive step holds.

By induction, we have constructed $g = b_0 + b_1x + b_2x^2 + \dots$ such that $fg = 1$. Therefore, f is a unit in $R[[x]]$.

- (b) Let f be a nilpotent element such that $f^n = 0$ for some $n \in \mathbb{N}$. We aim to show that all coefficients of f are nilpotent.

Base Case: Since $f^n = 0$, the constant term $a_0^n = 0$. Hence, a_0 is nilpotent.

Inductive Step: Assume that all coefficients $a_0, a_1, \dots, a_{n-1}$ are nilpotent. Consider the coefficient a_n .

Since $a_0, a_1x, \dots, a_{n-1}x^{n-1}$ are nilpotent by the inductive hypothesis, it follows that their sum, $a_0 + a_1x + \dots + a_{n-1}x^{n-1}$, is also nilpotent. This is because the sum of nilpotent elements is nilpotent.

Now, consider $f - (a_0 + a_1x + \dots + a_{n-1}x^{n-1})$. This simplifies to $a_nx^n + a_{n+1}x^{n+1} + \dots$. Since f is nilpotent, the difference $f - (a_0 + a_1x + \dots + a_{n-1}x^{n-1})$ is also nilpotent. Hence, there exists some $k \in \mathbb{N}$ such that

$$(f - (a_0 + a_1x + \dots + a_{n-1}x^{n-1}))^k = 0.$$

Expanding this expression, the leading term is $a_n^k x^{nk}$. Since the entire expression equals zero, it follows that $a_n^k = 0$. Therefore, a_n is nilpotent.

By induction, all coefficients of f are nilpotent.

The converse is not true.

- (c) Assume R is Noetherian. Let $\mathfrak{N}$ denote the ideal of nilpotent elements in R . By assumption, $\mathfrak{N} = (b_1, b_2, \dots, b_n)$, so every $b_1, \dots, b_n$ is nilpotent. Let m be the maximum power such that $b_i^m = 0$ for all i .

We aim to prove that $f = a_0 + a_1x + a_2x^2 + \dots \in R[[x]]$ is nilpotent by showing $f^{n(m-1)+1} = 0$.

Consider $f^{n(m-1)+1} = (a_0 + a_1x + a_2x^2 + \dots)^{n(m-1)+1}$. A general term in this expansion contributing to the coefficient of x^k is of the form:

$$a_{i_1}a_{i_2}\dots a_{i_t}, \quad \text{where } i_1 + i_2 + \dots + i_t = k \quad \text{and } t = n(m-1) + 1.$$

Each coefficient a_{i_j} is nilpotent, so $a_{i_j} \in \mathfrak{N} = (b_1, b_2, \dots, b_n)$. Thus, we can write:

$$a_{i_j} = r_{i_j,1}b_1 + r_{i_j,2}b_2 + \dots + r_{i_j,n}b_n, \quad \text{with } r_{i_j,k} \in R.$$

Expanding the product $a_{i_1} a_{i_2} \dots a_{i_t}$, each term is a product of the form:

$$r b_1^{\alpha_1} b_2^{\alpha_2} \dots b_n^{\alpha_n}, \quad \text{where } r \in R \text{ and } \alpha_1 + \alpha_2 + \dots + \alpha_n = t.$$

Since $t = n(m-1) + 1$, by the pigeonhole principle, at least one $\alpha_i > m-1$. But $b_i^m = 0$, so any such term vanishes. Hence, every term in the expansion of $f^{n(m-1)+1}$ is zero.

Since every term in the expansion of $f^{n(m-1)+1}$ vanishes, we conclude that $f^{n(m-1)+1} = 0$. Thus, f is nilpotent.

Problem 6

(e) Since $\cos^2(x) + \sin^2(x) = 1$ this implies that $(\cos(x)) + (\sin(x)) = (1)$ thus we are done by problem 2.

(f) Consider T_i such that $T_i = (-\infty, 1/i]$. Thus we have:

$$T_1 \supsetneq T_2 \supsetneq T_3 \supsetneq \dots$$

By part (d), we know:

$$\mathcal{I}(T_1) \subsetneq \mathcal{I}(T_2) \subsetneq \mathcal{I}(T_3) \subsetneq \dots$$

These ideals are not equal because we can construct a function $f_n \in \mathcal{I}(T_n)$ such that:

$$f_n(x) = \begin{cases} 0 & \text{if } x \in T_n, \\ x - 1/n & \text{if } x \notin T_n. \end{cases}$$

Thus, f_n is zero for all of T_n , but it is not zero for T_{n+1} . Therefore, $\mathcal{I}(T_n) \subsetneq \mathcal{I}(T_{n+1})$.

Now consider the ideal $I = \bigcup_{n=1}^{\infty} \mathcal{I}(T_n)$. We show that I is indeed an ideal:

- Let $a, b \in I$. Then $a \in \mathcal{I}(T_k)$ and $b \in \mathcal{I}(T_m)$ for some $k, m \in \mathbb{N}$. Let $N = \max(k, m)$. Then $a, b \in \mathcal{I}(T_N)$, and since $\mathcal{I}(T_N)$ is an ideal, $a + b \in \mathcal{I}(T_N) \subseteq I$.
- Let $a \in I$. Then $a \in \mathcal{I}(T_k)$ for some $k \in \mathbb{N}$. For any $r \in R$, $ra \in \mathcal{I}(T_k) \subseteq I$.

Thus, I is an ideal.

Assume for contradiction that R is Noetherian. Then I must be finitely generated, say $I = (f_1, f_2, \dots, f_k)$ for some $k \in \mathbb{N}$. Each generator f_j must belong to some $\mathcal{I}(T_{i_j})$. Let $N = \max(i_1, i_2, \dots, i_k)$. Then $f_1, f_2, \dots, f_k \in \mathcal{I}(T_N)$, so $I \subseteq \mathcal{I}(T_N)$.

However, since $\mathcal{I}(T_N) \subsetneq \mathcal{I}(T_{N+1}) \subsetneq \dots$, we have $I \not\subseteq \mathcal{I}(T_N)$, which is a contradiction. Therefore, I is not finitely generated, and R is not Noetherian.